Test Case Matrix

Case-centric complement to TEST_OBJECTIVES.md (which is category-centric). One row per actual case — nothing here is a placeholder for a case that doesn't exist yet; see "Coverage Gaps" at the bottom for what's still unbuilt.

Column definitions:

- **1-9** — the TEST_OBJECTIVES.md categories (legend below). **X** means at least one EXAM.md question genuinely requires that category's specific investigative mechanics to answer — not just "the case happens to involve logs" or "one question generically asks if anything bad happened." E.g. a benign case's "is there evidence of an attack?" question doesn't earn an X in category 6 (Intrusion Path Identification) unless there's an actual technique/vector to reconstruct — there isn't, so it doesn't.
- **Tools Used** — what generated or supplied the case's evidence. Either EvidenceForge (version/commit noted, since not all cases have been reconfirmed against the live checkout — see ../AGENTS.md) or, for cases sourced from a real external corpus, the source and its SOURCES.md status.
- **Skills Enabled** — the specific Anthropic-Cybersecurity-Skills entries relevant to this case's questions. This is the library the AUT is actually enriched with (see TEST_OBJECTIVES.md's AUT-toolkit note) — not the other two repos surveyed there, which aren't adopted for AUT use.

Category legend: 1 Log Analysis · 2 Browser History · 3 Account Activity · 4 Network Connection History · 5 Timeline Reconstruction · 6 Intrusion Path ID · 7 Lateral Movement · 8 Data Exfiltration · 9 Report Generation

Matrix

Test Case	Status	1	2	3	4	5	6	7	8	9	Tools Used	Skills Enabled
ssh-shared-key-overlap	Active	X		X	X	X				X	EvidenceForge — version/commit not yet reconfirmed against the live checkout (predates it)	- analyzing-linux-audit-logs-for-intrusion - detecting-service-account-abuse
rdp-remote-file-write	Active	X		X	X	X					EvidenceForge v1.17.0, commit 567073b0	- extracting-windows-event-logs-artifacts - performing-log-analysis-for-forensic-investigation - building-super-timelines-with-plaso
windows-log-search-basics	Active	X									Real (not synthetic) data from JPCERTCC/log-analysis-training_v2, Hands-on/basis/ (SOURCES.md: Adopted). No incident narrative by design — pure log search/filtering test.	- extracting-windows-event-logs-artifacts - performing-log-analysis-for-forensic-investigation
windows-lateral-movement-ntds-exfil	Active	X		X	X	X	X	X	X	X	Real (not synthetic) data from JPCERTCC/log-analysis-training_v2, Hands-on/advance/ (SOURCES.md: Adopted). Four-host, six-stage intrusion narrative — every fact independently re-derived from converted/raw data, not the	- analyzing-cyber-kill-chain - detecting-service-account-abuse

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											source PDF's narrative (which itself contained two separate confirmed errors, resolved against raw data — see the case's `BRIEFING.md`). Richest single-case coverage in the matrix; only category 2 (Browser) is out of reach.	- `extracting-windows-event-logs-artifacts` - `performing-log-analysis-for-forensic-investigation`
`external-recon-no-breach`	Active	X			X		X			X	EvidenceForge v1.17.0, commit `567073b0`. First case built from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #9). No attacker storyline — a port scan gets denied except for two ports (one inert, one a single failed SSH login); tests restraint from the opposite direction of `ssh-shared-key-overlap` (network-recon noise vs. account-overlap noise). Caught and fixed a real generator-tooling bug (`port_scan` self-targeting its own segment silently produced zero evidence) and a `GROUND_TRUTH.md` accuracy defect independent of that bug — see the paired generator README.	- `performing-log-analysis-for-forensic-investigation` - `analyzing-cyber-kill-chain`
`credential-spray-domain-compromise`	Active	X		X		X	X	X		X	EvidenceForge v1.17.0, commit `567073b0`. Second case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #1). A spray-compromised employee account escalates via a Kerberoasting-consistent credential request to an over-privileged service account; tests whether the AUT separates "how they got in" from "what they compromised." The key discrimination signal (one attacker-authored Event 4648 among several legitimate ones for the same account) came from the engine's own baseline realism, not deliberate scenario design — verified directly against raw data before the exam was written around it.	- `analyzing-cyber-kill-chain` - `detecting-service-account-abuse` - `performing-log-analysis-for-forensic-investigation`
`insider-dns-tunnel-exfil`	Active	X		X	X				X	X	EvidenceForge v1.17.0, commit `567073b0`. Third case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #2). No attacker anywhere — a finance analyst with entirely legitimate access archives client files and exfiltrates them via a ~2-hour DNS-tunnel to a domain never otherwise seen in the environment. Tests discrimination against a decoy archiving event (unrelated legitimate backup) and restraint on the "unauthorized access" question (correct answer: no).	- `performing-log-analysis-for-forensic-investigation` - `extracting-windows-event-logs-artifacts`
`phishing-c2-beacon`	Active	X			X	X	X		X	X	EvidenceForge v1.17.0, commit `567073b0`. Fourth case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #3). Phishing attachment → macro-launched PowerShell → periodic HTTPS beacon; among 39 total connections to the C2	- `analyzing-cyber-kill-chain` - `performing-network-traffic-`

Test Case	Status	1	2	3	4	5	6	7	8	9	Tools Used	Skills Enabled
											host, one carries a manual attacker command distinguishable only by byte volume. A confirmed engine limitation (`process_ref`/`parent_ref` did not produce the declared parent-child lineage in rendered logs) meant Q2 had to be redesigned around timing/command-content correlation instead of process lineage — a deliberate, documented gap in the evidence, not a bug the exam papers over.	`analysis-with-zeek` - `extracting-windows-event-logs-artifacts`
`websqli-webshell-pivot`	Active	X			X	X	X	X		X	EvidenceForge v1.17.0, commit `567073b0`. Fifth case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #4). SQL injection against a public web app → webshell → pivot to an internal file server over an unrevoked legacy firewall exception. Both central discrimination signals (the real breach vs. ~255 automated scan requests; the real pivot vs. a routine database-backup connection to the same target) came from the engine's own baseline/scan realism, verified directly before the exam was written, not designed in.	- `analyzing-cyber-kill-chain` - `performing-network-traffic-analysis-with-zeek` - `performing-log-analysis-for-forensic-investigation`
`pth-lateral-logclear`	Active	X		X		X		X		X	EvidenceForge v1.17.0, commit `567073b0`. Sixth case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #5). A cloned local-admin credential is used to authenticate to three file servers within ~12 minutes, then the Security log on the middle host is cleared. A confirmed engine limitation (`AuthenticationPackageName` for network logons is a fixed 70/30 Kerberos/NTLM random roll, independent of account scope) forced the exam away from an "NTLM-where-Kerberos-expected" signal toward the account's cross-host identity/timing pattern instead. Also confirms `log_cleared` does not remove prior log content in this engine version — the exam tests whether the AUT verifies that rather than assumes it.	- `detecting-lateral-movement-in-network` - `performing-active-directory-compromise-investigation` - `extracting-windows-event-logs-artifacts`
`benign-breakglass-account`	Active	X		X		X		X		X	EvidenceForge v1.17.0, commit `567073b0`. Seventh case from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposal #6), the Windows/AD companion to `ssh-shared-key-overlap`. No attack at all — two on-call sysadmins share a documented break-glass account across two separate off-hours occasions; a third, unrelated employee's late-night activity is a deliberate distractor. Mirrors `ssh-shared-key-overlap`'s grading philosophy directly: penalizes both false-positive escalation and under-investigation equally.	- `detecting-service-account-abuse` - `performing-log-analysis-for-forensic-investigation` - `extracting-windows-event-logs-artifacts`

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dga-beacon-logclear	Active	X			X	X			X	X	EvidenceForge v1.17.0, commit 567073b0. Eighth case from TEST_EVIDENCEFORGE_PROPOSED_CASES.md (proposal #7). A masquerading executable drives a DGA domain-hunt (91 queries, 87 NXDOMAIN) before establishing a 5-hour beacon; tests whether the AUT characterizes the search pattern quantitatively rather than citing only the one domain that resolved. Second confirmation this session that log_cleared doesn't remove prior log content — the exam again tests verification over assumption.	- hunting-for-data-exfiltration-indicators - performing-network-traffic-analysis-with-zeek - extracting-windows-event-logs-artifacts
departing-employee-email-exfil	Active	X			X	X			X	X	EvidenceForge v1.17.0, commit 567073b0. Ninth case from TEST_EVIDENCEFORGE_PROPOSED_CASES.md (proposal #8). The simplest technical case in the set by design — no attacker, no exploit; a departing employee with entirely legitimate access emails three attachments to his personal address. A real Phase 2 finding (default outbound STARTTLS made every message, including the exfiltration emails, render with blank sender/recipient fields) meant the case was silently unanswerable until fixed. Tests report register/tone as much as fact-finding — over- or under-dramatizing the finding are both graded.	- hunting-for-data-exfiltration-indicators - performing-log-analysis-for-forensic-investigation
rogue-service-account-privcreep	Active	X		X		X		X		X	EvidenceForge v1.17.0, commit 567073b0. Tenth and final case from TEST_EVIDENCEFORGE_PROPOSED_CASES.md (proposal #10). A service account documented for one unattended automated purpose gets invoked interactively and used to add itself to Domain Admins; tests whether the AUT identifies the explicit-credentials event itself — not the later group-membership change — as the earliest point a control should have fired. The richest discrimination signal (22 legitimate credential-usage events across all four hosts, several using the same process the attacker used) came from the engine's own baseline realism; an independent audit caught the first answer-key draft understating host coverage and wrongly treating process name as reliable — SubjectUserName is the only fully reliable signal, fixed before shipping.	- detecting-service-account-abuse - performing-active-directory-compromise-investigation - extracting-windows-event-logs-artifacts

Note on the "negative result still counts" standard, established by a since-discarded case: an earlier case (single-host-linux-rce, pulled from active use — see TEST_CASE_PROCESS.md Phase 6 for the discard criteria) had a lateral-movement question (Q10) requiring the AUT to check specific cross-host/cross-account evidence (WS-OP-01, sam.ortiz) for pivot indicators and correctly conclude none existed. That "checked a specific path and ruled it out" standard — as opposed to a generic "was anything malicious happening" catch-all with no path to trace — is the bar the two notes below apply to external-recon-no-breach and insider-dns-tunnel-exfil. That case's own row and record no longer exist in this table or in forensic-agent-tests; its full discard writeup briefly lived in this repo's _discarded/case-single-host-linux-rce/ before being deleted in an unrelated repo-reorg commit (6fecd3227, 2026-08-28) — recoverable from git history if ever needed, not duplicated here.

Note on `external-recon-no-breach`'s 6 (Intrusion Path ID): its X is deliberately a *negative* result, not a catch-all question — Q5 specifically requires checking multiple named hosts individually for any sign of compromise/follow-on activity before concluding no intrusion path exists, the same standard the discarded row's 7 note above holds lateral-movement checks to. This is different from a case that never asks the question at all.

Note on `insider-dns-tunnel-exfil`'s 3 (Account Activity): its X is also a negative result — Q4 requires citing the specific Event ID, Logon Type, and source field to justify "no unauthorized access," not a generic "nothing seemed wrong." Same standard as above.

Note on `benign-breakglass-account`'s 7 (Lateral Movement): category 7's own definition in `TEST_OBJECTIVES.md` explicitly covers "in a benign case, a legitimate user working across systems" — Q1 requires tracing the shared account across two host pairs (`APP-01`→`DB-01`, `FILE-01`→`DC-01`) and correctly grouping them into occasions, a real (if benign) cross-host correlation task, not a catch-all.

Coverage Gaps

Among **active** cases only:

- **Category 2 (Browser History)** — zero coverage anywhere, active or discarded. Known deficiency (see `TEST_OBJECTIVES.md`); no tool in the current pipeline produces browser artifacts. The only category with no identified near-term path to coverage, and now the *only* structural gap left in the entire matrix.

Categories 1, 3, 4, 5, 6, 7, 8, and 9 are now all covered by multiple active cases apiece — as of the eight cases added from `TEST_EVIDENCEFORGE_PROPOSED_CASES.md` (proposals #2-#8 and #10; #1 and #9 were built earlier in the same effort), category 9 (Report Generation) in particular went from "only two cases touch it generically" to real, question-level report/recommendation/register tests in most of the newly-built cases (`insider-dns-tunnel-exfil`, `phishing-c2-beacon`, `websqli-webshell-pivot`, `pth-lateral-logclear`, `benign-breakglass-account`, `dga-beacon-logclear`, `departing-employee-email-exfil`, `rogue-service-account-privcreep`), closing the "no case dedicated to report generation" gap this section previously flagged. `departing-employee-email-exfil` is the closest thing to a case *dedicated* to category 9 — its central test is report register/tone, not investigative depth.